ISLAMIC UNIVERSITY OF TECHNOLOGY

Course Outline and Course Plan

Name of the Teacher	Dr. Abu Raihan Mostofa Kamal	**Position**	Professor
Department	Computer Science and Engineering	**Programme**	B.Sc. Engineering in CSE / B.Sc TE (TVE)
Course Code	CSE 4307 / CSE 4173	**Course Title**	Database Management Systems
Academic Year	2023-2024	**Semester**	Winter
Contact Hours	3	**Credit Hours**	3.00
Text books and Reference books (if any)	Text book: 1. Database System Concepts, 7th edition, Silberschatz, Korth, and Sudarshan		
Prerequisites (If any)		**Course Type**	Compulsory
Course Homepage	Google Classroom Code: **vzv5swg**		
Teaching Methods/ Approaches	Lecture / Problem solving		

Course Assessment Method						
Attendance (10%)	**Quiz 30% of Total Marks (Best 3 out of 4)**				**Mid Semester (25%)**	**Semester Final (35%)**
	1st Quiz	**2nd Quiz**	**3rd Quiz**	**4th Quiz**	**Week/Date**	**Week/Date**
	Week/Date	**Week/Date**	**Week/Date**	**Week/Date**		
	4th week	8th week	11th week	13th week	As scheduled by IUT	As scheduled by IUT

Course Objectives	The courses aims for students to achieve the followings: 1) Understand and explain the basic principles of relational database management systems 2) Design the Entity-Relationship Models from a real-life high- level system description 3) Interact with database management systems by means of their languages (in particular SQL), and other interfaces, such as JDBC 4) Build and tune database application programs

Mapping with CO, PO and Bloom's Taxonomy			
CO No.	**Course Outcomes (CO) Statement**	**Levels of Bloom's Taxonomy**	**Matching with Program Outcome (PO)**
CO1	Understand the basic concepts of modern database management systems	C3	PO1
CO2	Design standard database applications from a higher level description	C4	PO3

CO3	Analyse and verify database design using standard techniques	C4	PO2

Weekly plan for course content		
Weeks	**Topics**	**Remarks**
1	Introduction to Database Systems: Evolution of file processing systems, role of databases in organizations, core components of a database environment;	
2	Data Modeling: the Entity- Relationship Diagram and its symbols and constructs;	
3	Transformation from high-level system description to entity-relationship model;	**Quiz 1 – CO1**
4	Transformation from high-level system description to entity-relationship model (cont.);	
5	SQL: basic structure, joins	**Quiz 2 – CO1**
6	SQL: aggregate functions, views, integrity constraints	
7	Secrutiy: user authorization, role based accress control, transformation of an entity-relationship data diagram into a relational model	
	Mid Term Exam	**CO1, CO2, CO3**
8	Database Interfacing: standard APIs (JDBC), SQL injections	
9	user-defined subprograms and triggers	**Quiz 3 – CO2**
10	Lossy and Lossless Decomposition, Functional dependencies	
11	Closure set and Armstrong's Axioms, Closure of Attributes	
12	Normalization theories (1NF, 2NF, 3NF, BCNF)	
13	Indexing: Badic Concept, primary and secondary index	**Quiz 4 – CO1, CO2**
14	Index Types: Dense and Sparse Index, Btree and B+ Tree	
15	B+ Tree Construction, Bitmap Indexing	
	Final Exam	**CO1,CO2, CO3**

Grading Policy

Numeric Grade	Letter Grade	Grade Point
80% and above	A+	4.00
75% to less than 80%	A	3.75
70% to less than 75%	A-	3.50
65% to less than 70%	B+	3.25
60% to less than 65%	B	3.00
55% to less than 60%	B-	2.75
50% to less than 55%	C+	2.50
45% to less than 50%	C	2.25
40% to less than 45%	D	2.00
Less than 40%	F	0.00

Class Schedule

Day		CSE Section B
Tuesday	10:30 – 11:45 (Sec 1)	14:30 – 15:45 (Sec 1)
Thursday	10:30 – 11:45 (Sec 2)	14:30 – 15:45 (Sec 1)

Student's consulting hour: Email for appointments.

Instructor contact details:

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Program Outcomes (POs : PO1 ~ PO12)

PO No.	Program Outcomes (POs)
	Students graduating from the Bachelor of Science in Civil Engineering (B.Sc. in Civil Engineering) program, upon graduation students will have the ability to:
PO1	**Engineering knowledge:** Apply knowledge of mathematics, natural science, engineering fundamentals and an engineering specialization as specified in K1 to K4 respectively to the solution of complex engineering problems.
PO2	**Problem analysis:** Identify, formulate, research literature and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences. (K1 to K4)
PO3	**Design/development of solutions**: Design solutions for complex engineering problems and design systems, components or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations. (K5)
PO4	**Investigation:** Conduct investigations of complex problems using research-based knowledge (K8) and research methods including design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.
PO5	**Modern tool usage:** Create, select and apply appropriate techniques, resources, and modern engineering and IT tools, including prediction and modelling, to complex engineering problems, with an understanding of the limitations. (K6)
PO6	**The engineer and society:** Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice and solutions to complex engineering problems. (K7)
PO7	**Environment and sustainability:** Understand and evaluate the sustainability and impact of professional engineering work in the solution of complex engineering problems in societal and environmental contexts. (K7)
PO8	**Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of engineering practice. (K7)
PO9	**Individual work and teamwork:** Function effectively as an individual, and as a member or leader in diverse teams and in multi-disciplinary settings.
PO10	**Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO11	**Project management and finance:** Demonstrate knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO12	**Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.